

Lab03_tavares

1.

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(stringr)
library(tidyr)

airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

— Column specification —

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
```

Rows: 322 Columns: 5

— Column specification —

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#a
colunas <- cols_only(
```

```

  DESTINATION_AIRPORT = col_character(),
  ARRIVAL_DELAY = col_double()
)

#b
linhasporvez <- 1000000

blocodeinfo <- function(chunk, pos) {
  chunk |>
    #c
    filter(!str_starts(DESTINATION_AIRPORT, "1")) |>
    #d
    drop_na(DESTINATION_AIRPORT, ARRIVAL_DELAY) |>
    #e
    group_by(DESTINATION_AIRPORT) |>
    summarise(
      atrasosoma = sum(ARRIVAL_DELAY),
      total_voos = n(),
      .groups = "drop"
    )
}

estatisticas <- read_csv_chunked(
  file = "flights.csv.zip",
  callback = DataFrameCallback$new(blocodeinfo),
  chunk_size = linhasporvez,
  col_types = colunas
)

#f
flights <- estatisticas |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    ARRIVAL_DELAY = sum(atrasosoma) / sum(total_voos),
    .groups = "drop"
  )

```

2.

```

#a: DESTINATION_AIRPORT

#b: IATA_CODE

#c
aeroportos_ausentes <- flights |>
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

#d e e
flights <- flights |>
  left_join(

```

```
airports |> select(IATA_CODE, CITY, STATE),
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
```

3.

```
aeropertos_por_estado <- airports |>
  count(STATE, name = "num_aeropertos", sort = TRUE)
```

```
aeropertos_por_estado
```

```
# A tibble: 54 × 2
```

	STATE	num_aeropertos
	<chr>	<int>
1	TX	24
2	CA	22
3	AK	19
4	FL	17
5	MI	15
6	NY	14
7	CO	10
8	MN	8
9	MT	8
10	NC	8

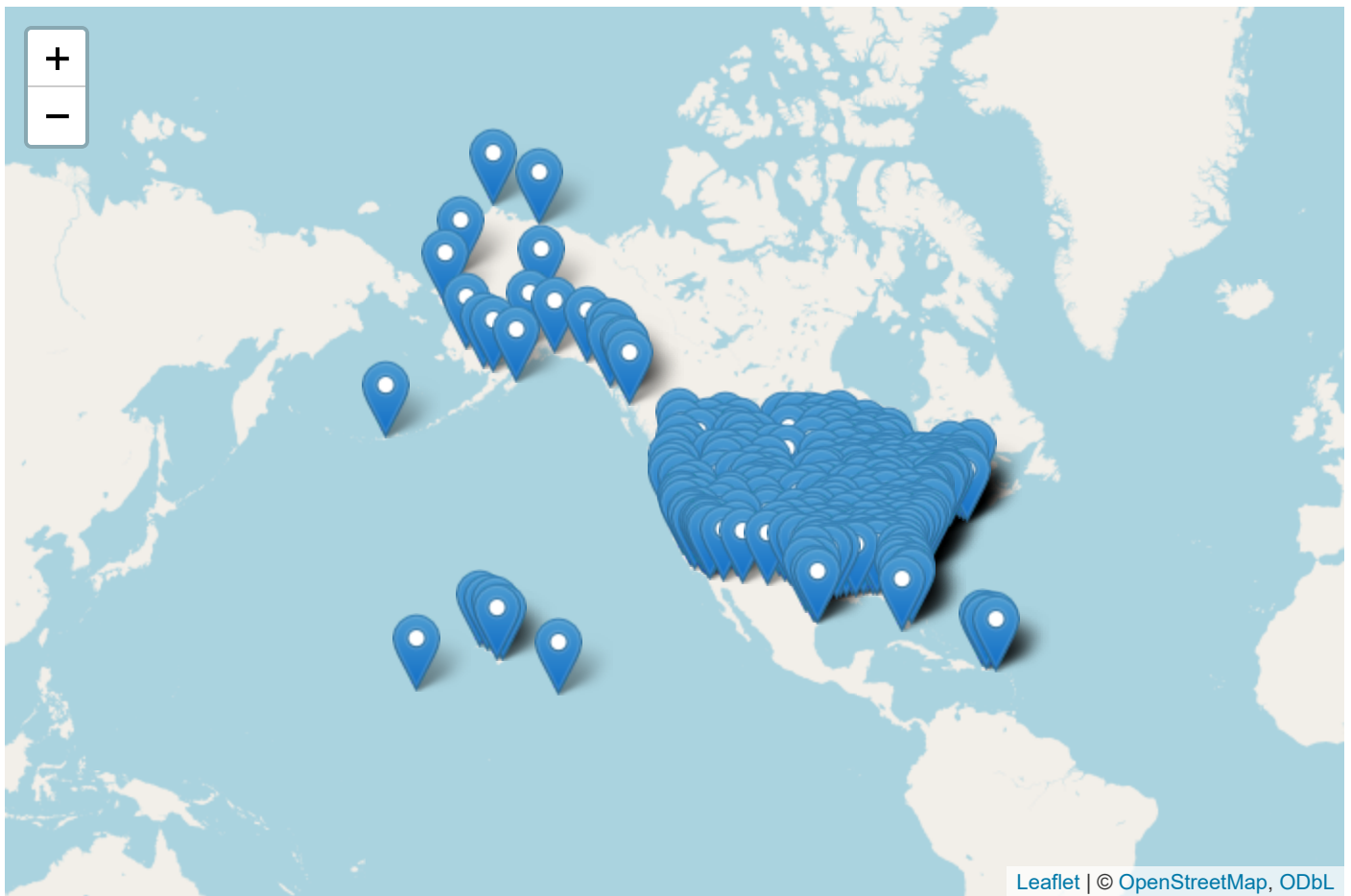
```
# i 44 more rows
```

4.

```
#a
library(leaflet)

# b e c
this <- flights |>
  left_join(
    airports |> select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) |>
  drop_na(LATITUDE, LONGITUDE) |>
  leaflet() |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0("<b>Aeroporto: </b>", DESTINATION_AIRPORT,
                    "<br><b>Atraso médio: </b>", round(ARRIVAL_DELAY, 2), " min")
  )

# d
this
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 19:14:17 -03"
```